Technology Stack

DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	**A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor**
MAXIMUM MARKS	4 MARKS

1. Introduction

The HDI Prediction System is built using a modern Python technology stack, utilizing lightweight libraries for data processing, machine learning model training, and web deployment.

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2. Technology Inventory

2.1 Core Development Environment

- **Language**: Python 3.12 (provides robust library support and standard performance).
- **IDE**: PyCharm (used for development, code inspection, and project management).

2.2 Machine Learning & Data Processing

- **Scikit-learn**:
- Used for training and evaluating the Linear Regression model (`sklearn.linear_model.LinearRegression`).
- Encodes country labels (`sklearn.preprocessing.LabelEncoder`).
- Serializes models (`pickle`).
- **Pandas & NumPy**:
- Pandas handles data loading, feature extraction, and preprocessing.
- NumPy manages matrix and array transformations for predictions.

2.3 Web Backend Layer

- **Flask**:
- A lightweight WSGI web framework.
- Routes requests and serves HTML templates.
- Integrates python predictive scripts directly with the client-facing front-end.
- **Jinja2**:
- Template rendering engine used by Flask.

- Renders predicted scores, color badges, progress bars, and insights dynamically.

2.4 Frontend Presentation Layer

- **HTML5 & CSS3**:
- Implements a modern dark-mode design system.
- Features glassmorphism cards with smooth backdrop blur filters and dynamic hover effects.
- Google Fonts (Inter) ensures clean typography.

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3. Technology Justification

Why Flask?

Unlike heavier frameworks like Django, Flask is lightweight and ideal for data science prototypes. It allows developers to deploy ML models with minimal boilerplate code while providing full routing flexibility.

Why Linear Regression?

- **Interpretability**: Coefficients clearly indicate the relative impact of each feature (e.g., expected years of schooling has the highest impact).
- **Simplicity**: Fast training times, low memory usage, and highly stable predictions.
- **Performance**: Achieves an $R^2 = 0.8834$ on our dataset, indicating a strong linear relationship.

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4. Software Architecture Layers

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│                      PRESENTATION LAYER                      │
│      - home.html (Dashboard Landing Page)                    │
│      - indexnew.html (Inputs and Result Rendering Card)      │
│      - style.css (Glassmorphic CSS Grid System)              │
└──────────────────────────────┬───────────────────────────────┘
                               │ HTTP GET/POST Requests
┌──────────────────────────────▼───────────────────────────────┐
│                    APPLICATION CONTROLLER                     │
│      - Flask Router (app.py)                                 │
│      - Route: / (Serves landing page)                        │
│      - Route: /predict (GET inputs / POST predictions)       │
│      - classify_hdi() & get_hdi_insights()                   │
└──────────────────────────────┬───────────────────────────────┘
                               │ Loads Serialized Artifacts
┌──────────────────────────────▼───────────────────────────────┐
│                    MACHINE LEARNING ENGINE                   │
│      - models/hdi_model.pkl (Linear Regression model)        │
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│       - models/encoder.pkl (LabelEncoder)                          │
│       - Preprocessing module (data_preprocessing.py)               │
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                                   │ Input Data Source
┌──────────────────────────────────▼─────────────────────────────────┐
│                             DATA LAYER                             │
│       - data/hdi_data.csv (200 records, 5 countries)               │
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